

COVID-19—A Reference List and Visualized Qualitative Aviation Research Themes

Chien-tsung Lu

School of Aviation & Transportation Technology
Purdue University
West Lafayette, IN 47906, USA
ctl@purdue.edu

Huabo Sun

Institute of Aviation Safety
China Academy of Civil Aviation Science and Technology
Beijing, China, 100028
sunhb@castc.org.cn

Abstract—The purpose of this study is to review pandemics-related publications that help aviation industry cope with pandemics outbreak like COVID-19. Published documents were searched and downloaded from academic libraries including Web of Science for a qualitative analysis. A reference list extracted from governmental publications for decision-makers, researchers, and program executives was provided. Important research clusters were visually generated based on VOSviewer process. Some research clusters were further discussed for a thorough understanding of the existing research perspectives.

Keywords—COVID-19, aviation emergency response, emergency preparedness, qualitative data visualization

I. INTRODUCTION

The outbreak of COVID-19 in the early 2020 has caused a substantial economic and financial damage to global systems, including air transportation industry. The COVID-19 has been announced by the World Health Organization (WHO) as a human-to-human communicable disease affecting the public health globally [1]. The damage has been nearly catastrophic as most airlines have decided to ground a large portion of fleet due to the lack of passengers as well as governmental restrictions on air travelers. Simultaneously, the massive reduction of passengers has directly affected airport's operation and revenue. Some of the impact resulted from an ineffective response to the disease spread throughout the world via components of air transportation systems. According to a report published in June 2020 by the International Air Transport Association (IATA), the COVID-19 has caused impact far severe than previous crises such as Global Financial Crisis (2007-2008) [11], 9/11 terrorist (2001) and SARS (2002-2003). A recent report published by International Civil Aviation Organization (ICAO) on February 10, 2021 also echoes that the COVID-19 pandemic outbreak has resulted in more than 60% (1,699 million) passenger volume reduction worldwide and financial loss was close to USD\$371B [12][13]. While the U.S. airline market has encountered financial loss close to USD\$88B by February 2021, the *Coronavirus Aid, Relief and Economic Security* (CARES) Act of 2020 had enabled federal government to allocate USD\$10B to eligible airports and USD\$25B to airlines (with additional USD\$25B for loans) to help aviation industry survive [5] followed by the passage of *Coronavirus Response and Relief Supplemental Appropriation Act* (CRRSAA) [3]. The entire industry is working diligently to combat pandemic spread, yet the virus is evolving into variants. In other words, fighting virus would need a long-term effort that governments and aviation

industry need to learn from each other.

It would be beneficial to aviation community and decision-makers if a list of leading governmental publications or alike can be prepared for a quick reference. It is also critical to identify specific topics and perspectives of existing research projects related to COVID-19, which shows key highlights of the documents. Simultaneously, the purposes of this study are threefold: (1) discovering governmental documentations, (2) finding existing research themes and (3) identifying specific preventive strategies counteracting COVID-19 outbreak or transmission.

II. LITERATURE REVIEWS

A. Economic Impact Studies

The COVID-19 has not only caused illnesses ranging widely but also imposed limitations on the global economy in different industries such as petroleum, stock market, tourism, hotels, retailing and regular societal commerce just to name a few. Economists like Hutt [10] of the World Economic Forum (WEF) noted that a slowdown of global economic growth was unavoidable, and the world would face a greater recession because of the COVID-19 outbreaks [10]. While airports serve as intersections of aviation systems, it was in 2011, Nishiura and Kamiya (2011) tested fever screening process at an airport during a pandemic event at a specific international airport. Their study focused quantitatively on the feasibility of a screening method or strategy during the pandemic outbreak [15]. In addition, in a TRB report, Smith & Greenberg (2017) conducted six case studies to measure the quality of airport emergency response plans and mitigation strategies related to two pandemic outbreaks - SARS in 2003 and Ebola in 2014 [17]. Chung (2015) suggested that less severe impact over airport economy could be achieved by a more efficient airport pandemic control plan and recommended a streamlined approach that would reduce the economic impact by improving overall effect of pandemic controls [4]. In the recent year, Gold and his research associates (2019) applied evaluation-by-simulation experiments to evaluate the effectiveness of disease screening strategies at airports [9]. All the aforementioned research projects have introduced preventive programs as well as recommendations for protection improvement.

B. Reports from Governmental Agencies

The World Health Organization (WHO) [20], U.S. Department of Transportation (DoT), Federal Aviation Administration (FAA), Center for Disease Control and

Prevention (CDC), Airport Council International (ACI)[1], Federal Emergency Management Administration (FEMA), Transportation Security Administration (TSA), Government Accountability Office (GAO) and many others had worked persistently to develop preventive and recovery programs aiming to provide guidelines to the public as well as the air operators to prevent further outbreak of coronavirus. The pandemic prevention program is not new to the industry as TSA has collaborated with CDC to educate passengers, airlines and airports through various workshops, forums, publications or media channels that helped properly cope with communicable diseases such as SARS, Ebola, H1N1, Swine Flu, etc. In fact, according to FAA AC 150/5200-37 Chapter 6 Section 8, airports' emergency response plans shall include "emergency medical service (EMS), public health, environmental health, mental health, and mortuary services" [6] and many scheduled airport emergency response drills or practices of emergency programs shall be conducted to respond to related undesired events. It is obvious that before COVID-19, most anticipated undesired events were related to aircraft incidents, accidents or traditional on-board medical emergencies instead of epidemic or etiological threats. To reduce the risk of spreading quarantinable diseases via air travel, aviation stakeholders need to practice well to mitigate identified communicable diseases. A quick reference responding to pandemic outbreak is imperative.

C. Aviation Safety Management System (SMS)

Identifying potential hazards affecting aviation operations is the fundamental concept of the contemporary risk management. The assurance of aviation safety has evolved from a reactive to a proactive and predictive fashion. The rationality of a proactive and predictive safety program is to detect hazards or threats and mitigate them before resulting in an accident. To be more proactive in promoting aviation safety, in 2000, Safety Management Systems (SMS) was introduced by the FAA in the U.S. and has increasingly gained recognition from the industry. In 2010, the *Airline Safety and Federal Aviation Administration Extension Act 2010* was passed to promote SMS implementation. On October 7, 2010, the FAA announced a new Notice of Proposed Rulemaking (NPRM) to collect comments from the industry regarding a mandatory airport SMS. As the aviation industry has anticipated, the new SMS regulation for Part 121 aviation service providers has been published as 14 CFR Part 5 [8].

D. Safety Risk Management (SRM)

There are desires to understand how safety risk management (SRM) is used to decide the level of severity and likelihood of occurrence for decision makers. Since August 30, 2010, the FAA Order 5200.11 has started to mandate airport SRM noting that from June 1, 2011 all categories of hub airports, from June 1, 2012 all FAR 139 airports, from June 1, 2013 all towered airports and from June 1, 2014 all NIPAS airports must conduct SRM [7]. Without doubt, epidemics-related risks should be no exception to this observation.

To be more proactive, Safety Risk Management (SRM) process is one of the four (4) fundamental pillars (policy, safety risk management, safety assurance and safety promotion) of SMS plan. SRM enables an organization to identify potential hazards, determine potential risks

(likelihood and severity), and select and implement appropriate risk mitigation strategies through a systemic manner [7]. It was in 2013, TRB Published "*Infectious Disease in Airports and on Aircraft*" aiming to provide guideline for strategies that can be implemented to mitigate communicable diseases [19]. While the FAA has recommended maintaining a correspondent hazard database including communicable diseases for risk assessment and assurance, the list of related references is useful to the stakeholders when encountering epidemic hazards. In the same vein, if the epidemic hazard collection tool is nonexistent, assessment could not be followed or calculated accurately. The result could mislead decision-makers in program design or improvement. Equally important, a visualized research focuses/themes could provide readers and researchers a holistic view regarding what critical perspectives or areas in counteracting pandemics and communicable diseases.

E. Research Questions

- 1) What leading pandemic-related publications from governments are available to aviation community?
- 2) What are the specific topics and perspectives of existing aviation research projects related to COVID-19?
- 3) What specific strategies were studied to prevent COVID-19 outbreak or transmission?

III. RESEARCH APPROACHES AND METHODOLOGY

To answer Question 1, this paper began with the search on academic databases that contain government's published papers, manuals and guidelines from which the researchers were able to download. For Questions 2 and 3, some paired keywords for publication searches were as the following: (1) "COVID-19" and "Aviation Industry", (2) "COVID-19" and "Air Transportation", (3) "COVID-19" and "Aviation", (4) "Coronavirus" and "Aviation Industry", (5) "Coronavirus" and "Air Transportation", (6) "Coronavirus" and "Aviation" and "COVID-19" and "Controls." Accordingly, seven (7) qualitative datasets were downloaded.

Downloaded documents were categorized for a quick reference, while for Questions 2 and 3, Web of Science was used to download publications in the full-record format. These downloaded documents were then converted into .txt format for VOSviewer process. VOSviewer is a program that uses Artificial Intelligence (AI) to review, analyze and extract qualitative themes, highlights or clusters available to researchers for a further interpretation.

A. Research Approach

The study started with the search on published literatures to identify manuals, guidelines, regulations and the like related to etiological trace and controls, health screening, hygiene, and infection protection in the global aviation system. Documentation, manuals from government, member organizations, airport, airlines, consultation companies (CDC, WHO, ICAO, IATA, U.S. DoT, EASA, FAA, TSA, ACI, ALPA, GAO, TRB etc.) and many others were categorized as a reference list for future emergency response program design and development such as IATA's quick template for emergency response plan [11]. While aviation is global in nature, international sources from foreign representative nations were also collected. The literature

triangulation process has been indispensable because the refined lessons learned or successful experiences from the global aviation industry are of great interest to air operators as well as decision-makers. Table I shows the tasks of this research project.

TABLE I TASKS OF THE PROJECT

Task No.	Required Tasks	Methodological Approach
1	Reference based on existing vetted materials (Q.1)	Documentation Reviews & Downloads (e-Libraries)
2	Converted .txt files and documents (Q. 2 and Q.3)	Web of Science
2	Visualized decision-making aids: themes, highlights or clusters (Q. 2 and Q.3)	VOSviewer and Meta-Analysis

B. Data Collection Exercise

To ensure a detailed analytical insight, the researchers applied a mix of documentation reviews, qualitative search using key words to generate a database including papers, best practices and guidelines ready for VOSviewer process. The researchers executed practical ideas and research initiatives using Action Research (AR) concept to conduct “look-think-act” process in order to form practical knowledge for the needed industry [2][16][18]. As Action Research (AR) concept is a scientific approach for evident discovery, the researchers experienced the first-hand challenges, process cognition and available knowledge, and implement selected strategies. A flow-chart provided below (Figure 1) is the AR procedure modified for this project.

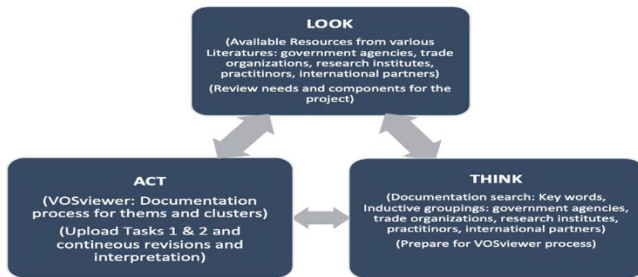


Fig.1 Action research execution flowchart

IV. FINDINGS

A. What leading pandemic-related publications from governments are available to aviation community?

The list of important governmental references is provided in Appendix. This final list was extracted and triangulated from the original 572 aviation-related publications (six keywords) and can be ready for further studies and provide researchers, aviation safety specialists and stakeholders a quick reference list for counteracting pandemic risks. During the study of this research question, Action Research’s “Look-Think-Act” process was considered and implemented to validate the list. All the keyword search results were compared, condensed, and prepared for VOSviewer’s qualitative visualization process, namely Question 2. To specify the criterion to answer Question 3, 229 highly cited papers were selected.

B. What are the specific topics and perspectives of existing aviation research projects related to COVID-19?

After the verification of the reference list, all identified

datasets were submitted to VOSviewer for a pictorial analysis. VOSviewer then generated visual interconnection maps and provided the following clustered research themes: COVID, pandemic, airline, passenger, country, and impact (see Figure 2). According to Figure 2, a further analysis showed that researchers were concerned about the impact from COVID-19 on both international and domestic airline passenger markets. The impact of pandemic outbreak on airports was also significantly discussed.

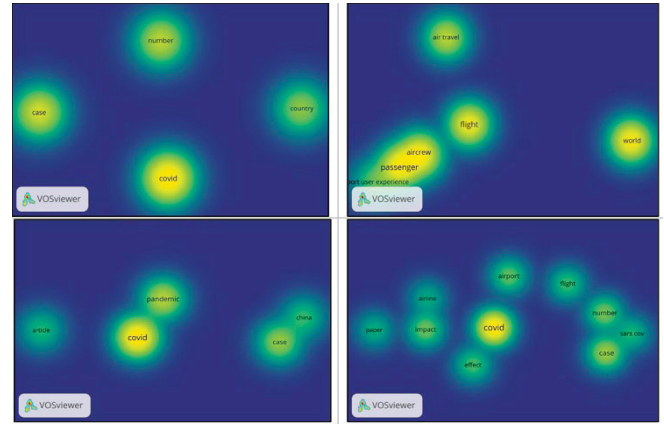


Fig.2 Individual Density Maps—Impact of COVID-19 upon Aviation Targets

The overall interconnection map is shown in Figure 3 including categorized COVID-related clusters (sector/India, time/year, passenger, case/number/data/country; aviation sector, impact & India; airline industry, research papers/articles; and pandemic study, airport, passenger and aviation industry) (See Figures 3). The setting of VOSviewer was based on “20” occurrences and “30” most relevant terms in this COVID-centered study. Of course, when using VOSviewer, the number of “occurrences” and “relevant terms” can be adjusted for a better visual demonstration according to the quantity of narrowed down documentations.

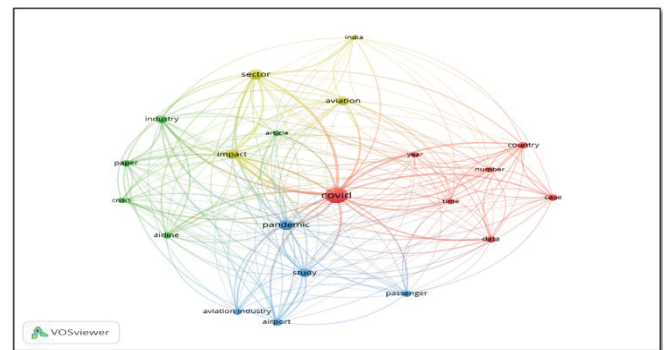
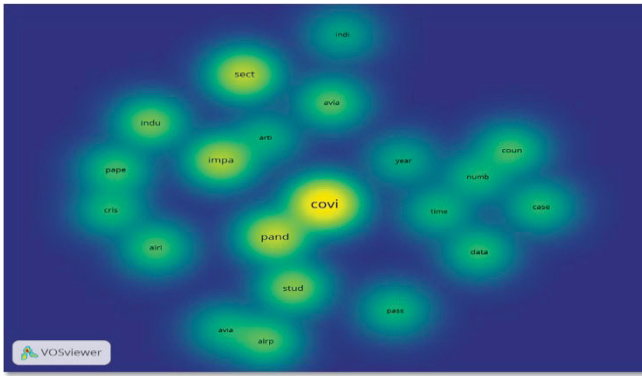
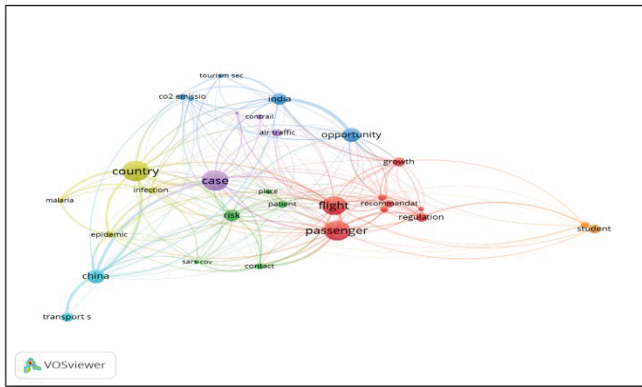


Fig.3 Overall Interconnection Map—COVID-19 Impact upon Aviation Targets

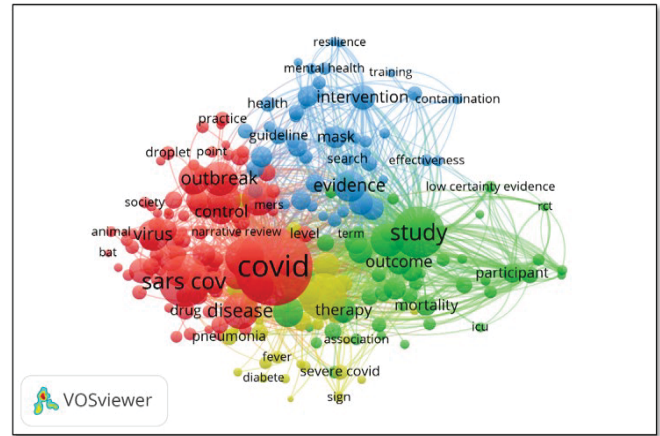
The density map shows below (Figure 4) reflecting on the clustered findings of Figure 3.



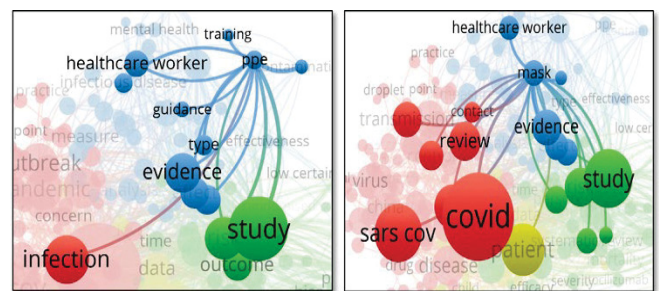
Important clustered finding was that China, India and Malaysia were three frequently mentioned countries. The clusters of “flight”, “passenger”, “case”, “opportunity” and “risk” were most studied (See Figures 5 & 6). It has been unique that “student” cluster was a stand-alone research theme and of course there were many known policies related to “students” and “COVID-19” during the unusual pandemic time.



C. What specific strategies were studied to prevent COVID-19 outbreak or transmission?



While other sub-clusters were also critically cited, such as *convalescent plasma*, *mask*, *child*, and *randomized controlled trial (RCT)*, this research revealed important approaches to control disease including *personal protective equipment (PPE)* and *Mask* (see Figure 9). Other clusters were: *vaccine development*, *elder adults and healthcare workers*, *anxiety and mental health*, *pathogen and human transmission*, *prevention and intervention*, *emergency response and modeling*.



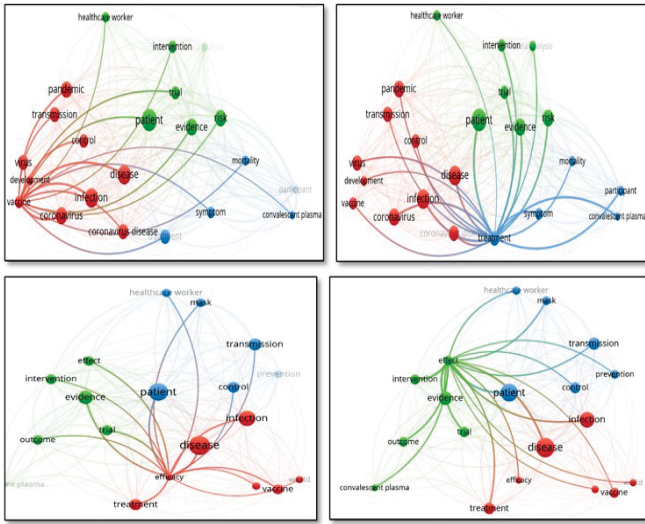


Fig.10 Selected COVID-19 Controls—Treatment, Vaccine, Efficacy & Effect

V. CONCLUSION

The result of this study created a reference list for the aviation stakeholders and managers counteracting the spread of communicable diseases not only in the U.S. but also the operators worldwide. This list helps aviation decision-makers proactively prevent business interruption due to pandemic outbreak as well as to generate practical recovery plans. The researchers applied Action Research (AR) concept and VOSviewer process to develop interconnection maps and clustered themes based on public publications in aviation fields. The VOSviewer categorized collected documents related to pandemic management and transformed them into visual landscapes that helps decision-makers and researchers to quickly identify perspectives that had been studied.

The governmental reference list was extracted and triangulated from 572 publications in relation to both “aviation” and “COVID-19.” For While Web of Science, it is comprehensive but only collects science-indexed publications for VOSviewer process. Thus, non-indexed research papers or articles were not reviewed by the researchers, which yields a follow-up study. The similar situation also applies to 229 most cited indexed papers involving keyword of “COVID-19” and “Controls”, published between 2020 and 2021. Future studies should focus on the trend of the pandemic preventive technologies.

ACKNOWLEDGMENT

This study is supported by the administrators of Purdue University IPG, SATT and Institute of Aviation Safety, China Academy of Civil Aviation Science and Technology. A special appreciation should be forwarded to the Organization Committee of ICCASIT 2021 for arranging this biannual international conference – Prof. Rongshun Huang, Prof. Jianya Gong, Prof. Ping Shu, Prof. Huabo Sun and Prof. Ruishan Sun for their continuous encouragement.

REFERENCES

[1] Airport Council International (ACI). (2020). The impact of COVID-19 on the airport business. <https://aci.aero/wp-content/uploads/2020/03/200401-COVID19-Economic-Impact-Bulletin-FINAL-1.pdf>

[2] Chalmers, H. & Colvin, J. (2005). Addressing environmental inequalities in U.K. policy: An Action Research perspective. *International Journal of Justice and Sustainability*, 10, 333-360. <https://doi.org/10.1080/13549830500160826>

[3] Coronavirus Response and Relief Supplemental Appropriation Act (CRRSAA) (Public Law 116- 260), H. R. 133, 116th Cong. (2021). <https://www.congress.gov/116/bills/hr133/BILLS-116hr133enr.pdf>

[4] Chung, L. H. (2015). Impact of pandemic control over airport economics: Reconciling public health with airport business through a streamlined approach in pandemic control. *Journal of Air Transport Management*, 44, 42-53.

[5] Coronavirus Aid, Relief, and Economic Security Act (CARES). (2020 March). H.R. 748.

[6] FAA. (2007). AC 150/5200-37C Instruction to safety management systems (SMS) for airport operators. Washington, D.C. Author.

[7] FAA. (2010 August). FAA Order 5200.11 FAA Airports (ARP) Safety Management System

[8] FAA. (2015, Jan. 8). FAA AC 120-92B System Safety Management for Aviation Service Providers. Washington, D.C.: Author.

[9] Gold, L., Balal, E., Horak, T., Cheu, R. L., Mehmetoglu, T., & Gurbuz, O. (2019). Health screening strategies for international air travelers during an epidemic or pandemic. *Journal of Air Transport Management*, 75, 27-38

[10] Hutt, R. (2020). The economic effects of the coronavirus around the world. Retrieved from <https://www.weforum.org/agenda/2020/02/coronavirus-economic-effects-global-economy-trade-travel/>

[11] International Air Transport Association (IATA). (2018). Emergency Response Plan: A template for Air Carriers. Retrieved from <https://www.iata.org/contentassets/f1163430bba94512a583eb6d6b24aa56/airlines-erp-checklist.pdf>

[12] IATA. (2020 June). COVID-19: Flexibility would be critical to success of the first year restart. Montreal, Canada. Author.

[13] CAO (2020) Economic impacts of COVID-19 on civil aviation. Retrieved from: <https://www.icao.int/sustainability/Pages/Economic-Impacts-of-COVID-19.aspx>

[14] ICAO. (2021 February). The effect of Novel Coronavirus (COVID-19) on civil aviation: the economic impact analysis. https://www.icao.int/sustainability/Documents/COVID-19/ICAO_Coronavirus_Econ_Impact.pdf

[15] Nishiura, H., & Kamiya, K. (2011). Fever screening during the influenza (H1N1-2009) pandemic at Narita International Airport, Japan. *BMC infectious diseases*, 11(1), 111.

[16] Lu, C-t., et al. (2011). System safety study: Pedagogical aviation Action Research. *JAAER*, 20(3). <https://doi.org/10.15394/jaer.2011.1641>

[17] Smith, J. F., & Greenberg, J. (2017). ACRP Synthesis 83: Preparing airports for communicable diseases on arriving flights: A synthesis of airport practices.

[18] Springer, E.T. (1996). Action research: A handbook for practitioners. Thousand Oaks, California. Sage.

[19] TRB ACRP. (2013, September 06) Report 91. Infectious Disease Mitigation in Airports and on Aircraft. http://onlinepubs.trb.org/onlinepubs/acrp/acrp_rpt_091.pdf

[20] World Health Organization (WHO). (2020). Coronavirus disease 2019 (COVID-19): situation report, 01. Retrieved from https://www.who.int/docs/default-source/coronavirus/situation-reports/20200121-sitrep-2019-ncov.pdf?sfvrsn=20a99c10_4

APPENDIX

[1] Australian Government Department of Health. (2021). Coronavirus (Covid-19) Advice for International Travelers. <https://www.health.gov.au/news/health-alerts/novel-coronavirus-2019-ncov-health-alert/coronavirus-covid-19-restrictions/coronavirus-covid-19-advice-for-international-travellers>

[2] Australian Department of Infrastructure, Transport, Regional Development and Communications. (2021, February 25). COVID-safe domestic flying – Domestic passenger journey protocol. <https://www.infrastructure.gov.au/aviation/covid-19-response/domestic-passenger-journey-protocol.aspx>

[3] Australian Representative on the Council of the ICAO. (2020, May 5). Adherence to ICAO Annex 9; and Actions taken to reduce the spread of Covid-19. <https://www.icao.int/Security/COVID-19/StateActions/Australia.pdf>

- [4] Canadian Air Transport Security Authority, CATSA. (2020). Measures taken in response to COVID-19. <https://www.catsa-acsta.gc.ca/en/covid19-response>
- [5] Public Health Agency Canada. (2020). Recommended disinfection and sanitization practices for aircrafts. <https://www.icao.int/APAC/COVID19%20BCP/Public%20Health%20Agency%20Canada%20-%20Recommended%20disinfection%20and%20sanitisation%20practices%20for%20aircrafts.pdf>
- [6] Civil Aviation Administration of China (CAAC). (2020). Technical Instructions on Disease Prevention, Control, and Protection for Civil Aviation Security Inspection Personnel (1st ed). Retrieved from <https://www.icao.int/APAC/COVID19%20BCP/Technical%20Instructions%201st%20edition.pdf>
- [7] CAAC. (2021). Preventing spread of coronavirus disease COVID-19 guideline for airports. CAAC. <https://www.icao.int/safety/CAPSCA/PublishingImages/Pages/Coronavirus/China-CAAC-PreventingSpreadofCOVID-19GuidelineAirports.pdf>
- [8] Civil Aviation Authority of Singapore. (2020, December 28). Health safety measures to mitigate risk of COVID-19 in aviation (Revision 1). <https://www.caas.gov.sg/docs/default-source/docs---spp/caas-document-on-health-safety-measures-for-covid-19.pdf>
- [9] Congressional Research Service (CRS). (2020, August 17). Addressing COVID-19 Pandemic Impacts on Civil Aviation Operations. <https://fas.org/sgp/crs/misc/R46483.pdf>
- [10] Department of Transportation (DoT). (2007, March 16). National Aviation Resource Manual for Quarantinable Diseases. Washington, D.C.: Author.
- [11] Departments of Transportation (DoT). (2020). *Runway to Recovery: The United States Framework for Airlines and Airports to Mitigate the Public Health Risks of Coronavirus*. Washington, D.C., https://www.transportation.gov/sites/dot.gov/files/2020-12/Runway_to_Recovery_1.1_DEC2020_Final.pdf
- [12] Department of Transportation. (2020, December 21). Runway to Recovery 1.1. U.S. <https://www.transportation.gov/briefing-room/runway-recovery-11>
- [13] Director General of Civil Aviation (Turkey) (2020). Airport Covid-19 Certification Guidelines. <http://web.shgm.gov.tr/documents/sivilhavacilik/files/Covid-19/DGCA-AIRPORT-GUIDELINES.pdf>
- [14] European Aviation Safety Administration (EASA). (2002). Quick risk assessment: Risk related to spread of new SARS-CoV2 variants of concern in the EU/EEA
- [15] EASA. (2020a) Operational Measures to Prevent the Spread of Coronavirus "SARS-CoV-2" Infection. https://www.easa.europa.eu/sites/default/files/dfu/easa_sd_2020_04.pdf
- [16] EASA. (2020b). Safety Information. Pitot-Static Issues After Storage Due to the COVID-19 Pandemic, 1-2. Retrieved March 14, 2021, from <https://www.easa.europa.eu/sites/default/files/dfu/sib-2020-14.pdf>
- [17] EASA. (2020c) Cabin Crew Recurrent Training Guidelines in relation to the COVID-19 pandemic https://www.easa.europa.eu/sites/default/files/dfu/easa_cc_recurrent_training_guidelines_covid19_issue_1_13nov2020.pdf
- [18] EASA. (2020d). Covid-19 aviation health safety protocol - operational guidelines for the management of air passengers and aviation personnel in relation to the covid-19 pandemic. https://www.ecdc.europa.eu/sites/default/files/documents/EASA-ECDC_COVID-19_Operational-guidelines-for-management-of-passengers-issue-2.pdf
- [19] EASA. (2020e). COVID-19 Resources. <https://www.easa.europa.eu/easa-covid-19-resources>
- [20] EASA. (2020, June 25). Operational measures to prevent the spread of coronavirus 'SARS-CoV-2' infection. SD No. 2020-04. https://www.easa.europa.eu/sites/default/files/dfu/easa_sd_2020_04.pdf
- [21] EASA. (2020, December 2). Guidelines for COVID-19 testing and quarantine of air travelers – Addendum to the aviation health safety protocol. https://www.easa.europa.eu/sites/default/files/dfu/guidelines_for_covid-19_testing_and_quarantine_of_air_travellers.pdfEASA. (2021).
- [22] European Centre for Disease Prevention and Control (2010). Risk assessment guidelines for diseases transmitted on aircraft. www.ecdc.europa.eu
- https://www.ecdc.europa.eu/sites/portal/files/media/en/publications/Publications/1012_GUI_RAGIDA_2.pdf
- [23] European Centre for Disease Prevention and Controls (2015, February). *Preparedness planning for respiratory viruses in EU Member States*. <https://www.ecdc.europa.eu/sites/default/files/media/en/publications/Publications/Preparedness%20planning%20against%20respiratory%20viruses%20-%20final.pdf>
- [24] European Commission (2020, March 26). Guidelines: Facilitating Air Cargo Operations during COVID-19 outbreak. Retrieved from <https://www.icao.int/APAC/COVID19%20BCP/European%20Commission%20Guidelines%20-%20Facilitating%20Air%20Cargo%20Operations%20during%20COVID-19%20outbreak.pdf>
- [25] Federal Aviation Administration (FAA). (2004, April). AC 150/5200-18 Airport safety self-inspection. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_150_5200-18C.pdf
- [26] FAA. (2009, June 19). Airport Emergency Plan. AC-150/5200-31C. https://www.faa.gov/documentLibrary/media/Advisory_Circular/150_5200_31c_consolidated.pdf
- [27] FAA. (2020a). Information for Airport Sponsors Considering COVID-19 Restrictions or Accommodations https://www.faa.gov/airports/airport_compliance/media/Information-for-Airport-Sponsors-COVID-19-Updated-22Dec2020.pdf
- [28] FAA. (2020b). Novel coronavirus: Interim health guidance from the federal aviation administration and the centers for disease control and prevention for air carriers and crews. <https://www.faa.gov/news/media/attachments/CDC%20FAA%20airline%20guidance.pdf>
- [29] FAA. (2020c). COVID-19 Vaccine Transport Considerations for Airport Operators. Washington, D.C.: Author. https://www.faa.gov/airports/special_programs/covid-19-airports/media/COVID-19-vaccine-transport-considerations.pdf
- [30] FAA. (2020d). Part 139 Cert Alert: Temporary Parking of Overflow Aircraft during COVID-19. https://www.faa.gov/airports/airport_safety/certalerts/media/part-139-cert-alert-20-02-COVID-19-temporary-aircraft-parking-updated-20200324.pdf
- [31] FAA. (2020e). Aviation Safety. Flight Standards Designee Oversight and Recurrent Training requirements related to the current Coronavirus (COVID-19) pandemic.
- [32] FAA. (2020f). Relief for Certain Persons and Operations During the Coronavirus Disease 2019 (COVID-19) Outbreak. <https://www.govinfo.gov/content/pkg/FR-2020-05-04/pdf/2020-09472.pdf>
- [33] FAA. (2020, March). Considerations for State, Local and Territorial COVID-19 Restrictions That Impact Air Transportation. https://www.faa.gov/airports/airport_compliance/media/State-Local-Territorial-Guidance-on-FAA. (2020, March 31). Air Traffic Control Operational Contingency Plan. Order JO-1900.47F. <https://www.skybrary.aero/bookshelf/books/5631.pdf>
- [34] FAA. (2020, April). AC 150/5200-37 Introduction to Safety Management Systems for Airport Operators. https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_150_5200-37.pdf
- [35] FAA. (2020, December). Information for airport sponsors considering covid-19 restrictions or accommodations. https://www.faa.gov/airports/airport_compliance/media/Information-for-Airport-Sponsors-COVID-19-Updated-22Dec2020.pdf
- [36] FAA. (2020, May 11). COVID-19 updated interim occupational health and safety guidance for air carriers and crews. SAFO 20009. <https://nbaa.org/wp-content/uploads/2020/04/faa-safo20009-1.pdf>
- [37] FAA. (2021, February 27). Coronavirus (COVID-19) Information from the FAA. <https://www.faa.gov/coronavirus/>
- [38] FAA. (2021, March 2). Airport Coronavirus Response Grant Program. Airport Coronavirus Response Grant Program – Airports. <https://www.faa.gov/airports/crrsaa/>
- [39] FAA. (2021, March 5). Coronavirus Guidance & Resources. https://www.faa.gov/coronavirus/guidance_resources/#gas
- [40] Federal Emergency Management Administration (FEMA). (2014). Pandemic Influenza Template. https://www.fema.gov/sites/default/files/2020-08/fema_pandemic-influenza_template-instructions.pdf

- [41] FEMA (n.d.). Coronavirus (COVID-19) response. <https://www.fema.gov/disaster/coronavirus>
- [42] Government Accountability Office (GAO) (2015). Subcommittee on aviation, Committee on Transportation and Infrastructure, House of Representatives, Air travel and communicable diseases, comprehensive federal plan needed for U.S. aviation system's preparedness. A Report to House of Representatives (2015). Washington, DC.
- [43] GAO. (2015, December). Air travel and communicable diseases: Comprehensive Federal Plan Needed for U.S. Aviation System's Preparedness. <https://www.gao.gov/assets/680/674224.pdf>
- [44] Government of Bangladesh. (2020, March). National Preparedness and Response Plan for COVID-19, Bangladesh. https://reliefweb.int/sites/reliefweb.int/files/resources/nprp_covid-19_v6_18032020.pdf
- [45] Government of Canada (2020, August). Canada's flight plan for navigating COVID-19. https://tc.canada.ca/sites/default/files/2020-08/CANADA-FLIGHT-PLAN_E-accessible.pdf
- [46] Government of Canada (2021). COVID-19: Guidance Material for Air Operators Managing Travelers to Canada at International Airports. Transport Canada. <https://tc.canada.ca/en/initiatives/covid-19-measures-updates-guidance-issued-transport-canada/air-operators-managing-travellers-during-check-procedure-international-airports>
- [47] Government of Japan (2020, March 28). Basic policies for novel coronavirus disease control by the Government of Japan (Revised 2020, May 25). <https://www.mhlw.go.jp/content/10900000/000634753.pdf>
- [48] Government in India (2021). Guidelines for international arrivals. <https://www.newdelhiairport.in/media/1622/latest-guidelines-for-international-arrivals.pdf>
- [49] House Hearing 109-9. (2005, April). gida.ghscosting.org. Effort to prevent pandemic by air travel. <https://gida.ghscosting.org/downloads/Health%20Security%20Net%20-%20Data%20Download.xlsx>
- [50] International Civil Aviation Organization (ICAO). (2008). Pandemic Preparedness Planning. <https://www.icao.int/Meetings/wrdss2011/Documents/DevelopmentForum2008/Singh.pdf>
- [51] ICAO. (n.d.). Global roadmap supporting the implementation of the recommendations and guidance of the council aviation recovery task force for the restart, recovery, and resilience of civil aviation following the COVID-19 outbreak. <https://www.icao.int/safety/CAPSCA/PublishingImages/Pages/Coronavirus/ICAO%20Global%20Implementation%20Roadmap%20of%20the%20Recommendations%20and%20Guidance%20of%20the%20Council%20Aviation%20Recovery%20Task%20Force%20-%20CART%209.pdf>
- [52] ICAO. (2014). Template for a National Aviation Public Health Emergency Preparedness Plan.
- [53] ICAO. (2015). Public Health Emergency Contingency Plan. <https://www.icao.int/Meetings/CAPSCA2015/Presentations/DAY%201/Session%201/04CAPSCAVisitsLessonsLearnedRev11.pdf>
- [54] ICAO. (2020a). Doc 10152: Manual on Testing and Cross-border Risk Management Measures. https://www.icao.int/covid/cart/Documents/Doc%2010152_Manual%20on%20Testing%20and%20Cross-border%20Risk%20Management%20Measures.pdf
- [55] ICAO. (2020b). Council Aviation Recovery Task Force Take-Off Guidance. <https://www.icao.int/covid/cart/Pages/Airports-Module.aspx>
- [56] ICAO. (2020c). Guideline for states concerning the management of communicable disease posing a serious public health risk. https://www.icao.int/safety/aviation-medicine/guidelines/avinfluenza_guidelines.pdf
- [57] ICAO. (2020d). Preventing spread of coronavirus disease 2019 (COVID-19) guideline for airlines [https://www.icao.int/Security/COVID-19/StateActions/Preventing%20Spread%20of%20Coronavirus%20Disease%202019%20\(COVID-19\)%20Guideline%20for%20Airports%204th.pdf](https://www.icao.int/Security/COVID-19/StateActions/Preventing%20Spread%20of%20Coronavirus%20Disease%202019%20(COVID-19)%20Guideline%20for%20Airports%204th.pdf)
- [58] ICAO. (2020e). Guidance on Economic and Financial Measures. https://www.icao.int/sustainability/Documents/COVID-19_Economic_and_Financial_Measures/ICAO_Guidance_on_Economic_and_Financial_Measures.pdf
- [59] ICAO. (2020f). ICAO Handbook for CAAs on the Management of Aviation Safety Risks related to Covid-19 (1st ed.). <https://www.icao.int/safety/SafetyManagement/Doc10144/Doc%2010144.pdf>
- [60] ICAO. (2020g). COVID-19 Response and Recovery Platform. <https://www.icao.int/covid/Pages/default.aspx>
- [61] ICAO. (2020h). Air Navigation Services (ANS) COVID-19 Recommendations. <https://www.icao.int/APAC/COVID19%20BCP/NACC82310ATM-States-ANS-COVID19Recommendations.pdf>
- [62] ICAO. (2020i). Implementation of contingency arrangements to reduce the risks of the spread of COVID-19. <https://www.icao.int/APAC/COVID19%20BCP/047c%20ICAOHQ%20-%20Implementation%20of%20contingency%20arrangements%20to%20reduce%20the%20risks%20of%20the%20spread%20of%20COVID-19.pdf>
- [63] ICAO (2021). Managing Communicable Disease in Aviation. <https://www.icao.int/safety/aviation-medicine/pages/healthrisks.aspx>
- [64] ICAO – CART. (2020, November 5). Take-off: Guidance for Air Travel through the COVID-19 Public Health Crisis. International Civil Aviation Organization – Council Aviation Recovery Task Force. <https://www.icao.int/APAC/COVID19%20BCP/SL20.67.EN.pdf>
- [65] Ministry of Health/Ghana Health Service (2020, August). Guidelines for International Flight Operations on COVID-19 Safety at the Kotoka International Airport. <http://www.gcaa.com.gh/web/?p=1031>
- [66] Ministry of Transport, Kenya (2020, June). Protocol for air travel operations during the COVID-19 public health crisis. https://kcaa.or.ke/sites/default/files/docs/covid_19/Protocol-for-Air-Operations.pdf
- [67] New South Wales Government (Australia). (2020, March). Aviation Emergency Sub Plan. <https://www.emergency.nsw.gov.au/Documents/plans/sub-plans/SubPlan-Aviation.pdf>
- [68] New Zealand Immigration. (2021). New Zealand border entry requirements. <https://www.immigration.govt.nz/about-us/covid-19/border-closures-and-exceptions/border-entry-requirements>
- [69] Nigerian Civil Aviation Authority (NCAA). (2020). *Post Covid-19 Guidance For Aerodrome Operators Resumption Of Operations*. Government of Nigeria. Retrieved March 14, 2021, <https://ncaa.gov.ng/media/0qapn5np/advisory-circular-ncaa-ac-adr-007-post-covid-19-aerodrome-restart-plan.pdf>
- [70] South African Government. (2021). Travel - Coronavirus COVID-19. https://www.gov.za/covid-19/travel-international?gclid=CjwKCAiAKjCBhAyEiwAKQBCKqh734Vcb-2_gIsles0XkWh-fr9JH9YXiRflnHUWfm9jDkBzpgQ_RoCE8gQAvD_BwE#
- [71] South Africa - Department of Transport. (2021, January 29). Disaster Management Act (57/2002: Measures to Prevent and Combat the spread of Covid-19 in the Air Services for Adjusted Alert Level 3, Government Gazette (44124). <http://www.caa.co.za/Legal%20Documents/Government%20Gazette%20Notice%20-%202029%20January%202021.pdf>
- [72] Taiwan Centers for Diseases Control (January 9, 2020). The emergency response to pandemic outbreak for the public transportation system.
- [73] Transport Canada (2021). COVID-19: Guidance material for air operators managing travelers during the check-in procedure at transborder (U.S.) airports.
- [74] Transport Canada. (2021). COVID-19 measures, updates, and guidance issued by Transport Canada <https://tc.canada.ca/en/initiatives/covid-19-measures-updates-guidance-issued-transport-canada>
- [75] Texas Department of Transportation. (2020, March 28). <https://ftp.txdot.gov/pub/txdot-info/avn/covid19/information-airport-sponsors-032820.pdf>
- [76] TRB Airport Cooperative Research Program (ACRP) (n.d.). Infectious disease resources. <http://www.trb.org/ACRP/ACRPInfectiousDiseaseResources.aspx>
- [77] TRB ACRP. (2006). *Quarantine stations at ports of entry: Protecting the Public's health*. <https://www.nap.edu/read/11435/chapter/3>
- [78] TRB ACRP. (2010, August 06). *Research on the transmission of disease in airports and on aircraft*. Retrieved March 14, 2021, from <https://www.nap.edu/catalog/22941/research-on-the>

transmission-of-disease-in-airports-and-on-aircraft

- [79] TRB ACRP. (2012). Report 74: Application of Enterprise Risk Management at Airports. DOI:10.17226/22744
- [80] TRB ACRP. (2012). Synthesis 37: Lessons Learned from Safety Management Systems Pilot Study. DOI:10.17226/22740
- [81] TRB ACRP. (2013, September 06) Report 91. Infectious Disease Mitigation in Airports and on Aircraft. http://onlinepubs.trb.org/onlinepubs/acrp/acrp_rpt_091.pdf
- [82] TRB ACRP. (2017a). Report 1: Safety Management Systems for Airports, Volume 2: Guidebook. DOI: 10.17226/14316
- [83] TRB. (2017b). Preparing Airports for Communicable Diseases on Arriving Flights (2017). DOI 10.17226/24880
- [84] TRB ACRP. (2018, December 11). Airport Response during Communicable Disease Outbreaks. <http://onlinepubs.trb.org/onlinepubs/webinars/181211.pdf>
- [85] TRB ACRP. (2019, January 23). *Airport roles in reducing transmission of communicable diseases*. Retrieved March 14, 2021, from <https://www.nap.edu/catalog/25367/airport-roles-in-reducing-transmission-of-communicable-diseases>
- [86] TRB ACRP. (2019, September 18). *Emergency working groups at airports*. Retrieved March 14, 2021, from <https://www.nap.edu/catalog/25572/emergency-working-groups-at-airports>
- [87] TRB. (2021, February 5). ACRP 04-25 (RFP): Airports and Communicable Diseases—Preparedness and Response. Retrieved from <https://apps.trb.org/cmsfeed/TRBNetProjectDisplay.asp?ProjectID=5030>
- [88] U.K. DoT. (2021, June 11). Coronavirus (COVID-19): safer aviation guidance for operators. <https://www.gov.uk/guidance/coronavirus-covid-19-safer-aviation-guidance-for-operators#at-the-airport>.
- [89] United Nations Humanitarian Air Service Republic of South Sudan (UNHAS). (2020, April 2). COVID19 Standard Operating Procedures (SOP). United Nations Humanitarian Air Service Republic of South Sudan. https://reliefweb.int/sites/reliefweb.int/files/resources/unhas_covid19_standard_operating_procedure_2_apr_2020.pdf
- [90] U.S. Centers for Disease Control and Prevention (CDC). (2005, May 03). Guidance for Airline Flight Crews, Cargo and Cleaning Personnel, and Personnel Interacting with Arriving Passengers. <https://www.cdc.gov/sars/travel/airpersonnel.html>
- [91] U.S. CDC. (2017). Guidance for Airlines on Reporting Onboard Deaths or Illnesses to CDC. Center for Disease Control and Prevention. <https://www.cdc.gov/quarantine/air/reporting-deaths-illness/guidance-reporting-onboard-deaths-illnesses.html>
- [92] U.S. CDC. (2019a). What airline workers need to know about COVID-19. <https://www.cdc.gov/coronavirus/2019-ncov/downloads/AirlineWorkers-ASSISTANCE.pdf>
- [93] U.S. CDC. (2019b). Preventing spread of disease on commercial aircraft: Guidance for cabin crew | quarantine. <https://www.cdc.gov/quarantine/air/managing-sick-travelers/commercial-aircraft/infection-control-cabin-crew.html>
- [94] U.S. CDC. (2020, March, 4). Updated interim guidance for airlines and airline crew: coronavirus disease 2019 (COVID-19). <https://www.cdc.gov/quarantine/air/managing-sick-travelers/ncov-airlines.html>
- [95] U.S. CDC. (2021, January 29). Requirement for persons to wear masks while on conveyances and at transportation hubs. https://www.cdc.gov/quarantine/pdf/Mask-Order-CDC_GMTF_01-29-21-p.pdf
- [96] U.S. CDC. (2021, February 16). Travel During COVID-19. <https://www.cdc.gov/coronavirus/2019-ncov/travelers/travel-during-covid19.html>
- [97] U.S. CDC. (2021, February 18). Testing and International Air Travel. <https://www.cdc.gov/coronavirus/2019-ncov/travelers/testing-air-travel.html>
- [98] U.S. CDC. (2021, June 11.). COVID-19: Airport Passenger Assistance Workers. <https://www.cdc.gov/coronavirus/2019-ncov/community/organizations/airport-passenger-assistance-workers.html>
- [99] U.S. Environmental Protection Agency (EPA) (n.d.). List N Tool: COVID-19 Disinfectants. United States Environmental Protection Agency. <https://cfpub.epa.gov/giwiz/disinfectants/index.cfm>
- [100] U.S. National Academies of Sciences, Engineering, and Medicine (Report 91) (2013). Infectious Disease Mitigation in Airports and on Aircraft. Washington. The National Academies Press. <https://doi.org/10.17226/22512>
- [101] U.S. Department of Interior (DoI). (n.d.). Our response to COVID-19. <https://www.doi.gov/coronavirus>
- [102] U.S. Transportation Security Administration (TSA) (2020, May 21). TSA prepared for summer travelers with updated security procedures. <https://www.tsa.gov/news/press/releases/2020/05/21/tsa-prepared-summer-travelers-updated-security-procedures>.
- [103] U.S. TSA. (2021, March 8). Coronavirus (COVID-19) Information. <https://www.tsa.gov/coronavirus>
- [104] U.S. Consulate General Hong Kong & Macau. (2021). Covid-19 Information. <https://hk.usconsulate.gov/covid-19-information/>
- [105] U.S. Homeland Security Council. (2006, May). *National Strategy for Pandemic Influenza*. <https://www.cdc.gov/flu/pandemic-resources/pdf/pandemic-influenza-implementation.pdf>
- [106] U.S. Department of Homeland Security (DHS) (2020, November 14). New Qualification Program to Enhance Aviation Safety, COVID-19 Response. <https://www.hstoday.us/industry/new-qualification-program-to-enhance-aviation-safety-covid-19-response>
- [107] U.S. DHS. (2020, December 16). Advisory memorandum on ensuring essential critical infrastructure workers' ability to work during the COVID-19 response. https://www.cisa.gov/sites/default/files/publications/ECIW_4.0_Guidance_on_Essential_Critical_Infrastructure_Workers_Final3_508_0.pdf
- [108] U.S. Department of State (n.d.). Ethiopia, Revised Quarantine Requirements For Arriving Travelers. OSAC. <https://www.osac.gov/Country/Ethiopia/Content/Detail/Report/38fefe77-0e9b-4690-a4cd-19066d6bd848>.
- [109] U.S. Department of State - Bureau of Consular Affairs (2021). United Arab Emirates International Travel Information. <https://travel.state.gov/content/travel/en/international-travel/International-Travel-Country-Information-Pages/UnitedArabEmirates.html>
- [110] U.S. Departments of Transportation (DoT). (2020). Runway to recovery the United States framework for airlines and airports to mitigate the public health risks of coronavirus. https://www.transportation.gov/sites/dot.gov/files/2020-07/Runway_to_Recovery_07022020.pdf
- [111] World Health Organization (WHO). (2015). Handbook for the Management of Public Health Events in Air Transport. https://www.who.int/ihr/publications/9789241510165_eng/en/.
- [112] WHO. (2020a). Management of ill travelers at Points of Entry (international airports, seaports, and ground crossings) in the context of COVID-19. <https://apps.who.int/iris/bitstream/handle/10665/331512/WHO-2019-nCoV-POEmgmt-2020.2-eng.pdf>
- [113] WHO. (2020b). Operational considerations for managing COVID-19 cases or outbreak in aviation: interim guidance. <https://www.who.int/publications/i/item/operational-considerations-for-managing-covid-19-cases-or-outbreak-in-aviation-interim-guidance>
- [114] WHO. (2020, March 18). Operational considerations for managing COVID-19 cases or outbreak in aviation. <https://apps.who.int/iris/bitstream/handle/10665/331488/WHO-2019-nCoV-Aviation-2020.1-eng.pdf>
- [115] WHO. (2020, November 4). Critical preparedness, readiness and response actions for COVID-19. <https://www.who.int/publications/i/item/critical-preparedness-readiness-and-response-actions-for-covid-19>